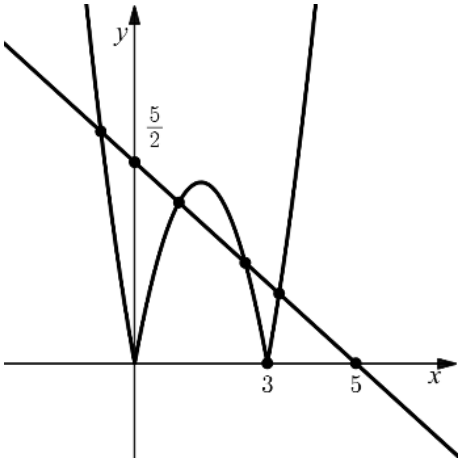
1) a)



1) b)

$x^2 - 3x = \frac{5}{2} - \frac{1}{2}x \Rightarrow 2x^2 - 5x - 5 = 0$

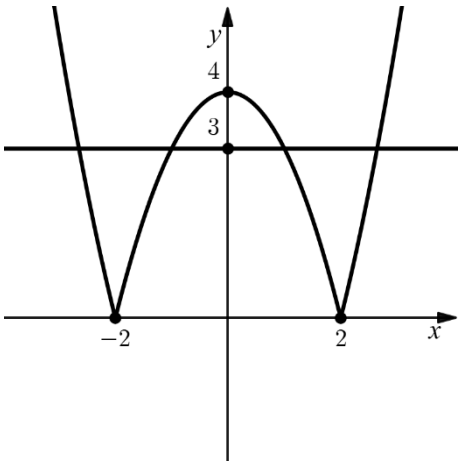
$x = \frac{5}{4} \pm \frac{\sqrt{65}}{4}$

$x^2 - 3x = -\frac{5}{2} + \frac{1}{2}x \Rightarrow 2x^2 - 7x + 5 = 0$

$x = 1$ or $x = \frac{5}{2}$

$\therefore x \in \left(-\infty, \frac{5}{4} - \frac{\sqrt{65}}{4}\right) \cup \left(1, \frac{5}{2}\right) \cup \left(\frac{5}{4} + \frac{\sqrt{65}}{4}, \infty\right)$

2) a)

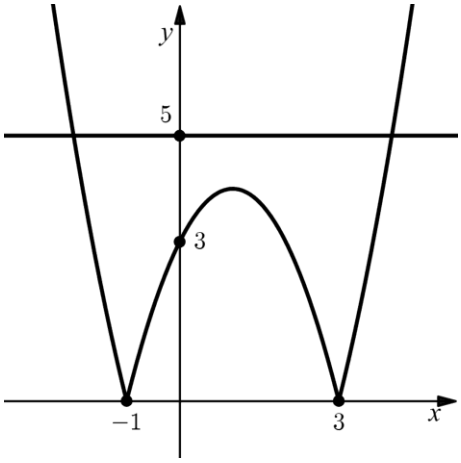


$x^2 - 4 = 3 \Rightarrow x^2 = 7$ so $x = -\sqrt{7}$ or $x = \sqrt{7}$

$x^2 - 4 = -3 \Rightarrow x^2 = 1$ so $x = -1$ or $x = 1$

$\therefore \{x: x \in \mathbb{R}, x < -\sqrt{7}\} \cup \{x: x \in \mathbb{R}, -1 < x < 1\} \cup \{x: x \in \mathbb{R}, x > \sqrt{7}\}$

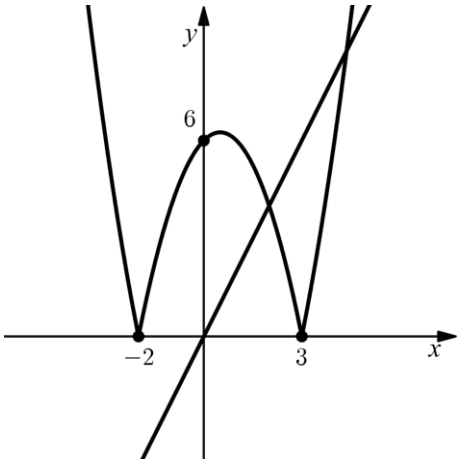
2) b)



$x^2 - 2x - 3 = 5 \Rightarrow x^2 - 2x - 8 = 0$ so $x = -2$ or $x = 4$

$\therefore \{x: x \in \mathbb{R}, -2 \leq x \leq 4\}$

2) c)



$-(x^2 - x - 6) = 2x \Rightarrow x^2 + x - 6 = 0$ so $x = -3$ or $x = 2$

$x^2 - x - 6 = 2x \Rightarrow x^2 - 3x - 6 = 0$ so $x = \frac{3}{2} - \frac{\sqrt{33}}{2}$ or $x = \frac{3}{2} + \frac{\sqrt{33}}{2}$

$\therefore \{x: x \in \mathbb{R}, x < 2\} \cup \left\{x: x \in \mathbb{R}, x > \frac{3}{2} + \frac{\sqrt{33}}{2}\right\}$

3) a)

$k(x + 3)(x - 2)$ so $k = -2$

$\pm 2(x + 3)(x - 2) = \pm(2x^2 + 2x - 12)$

$\therefore a = 2$, $b = 2$ and $c = -12$ or $a = -2$, $b = -2$ and $c = 12$

3) b)

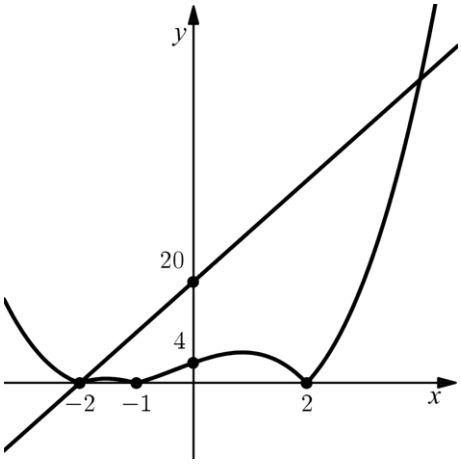
$2x^2 + 2x - 12 = x \Rightarrow\ 2x^2 + x - 12 = 0 \therefore x = -\frac{1}{4} \pm \frac{\sqrt{97}}{4}$

$2x^2 + 2x - 12 = -x \Rightarrow\ 2x^2 + 3x - 12 = 0 \therefore x = -\frac{3}{4} \pm \frac{\sqrt{105}}{4}$

$\therefore x = -\frac{1}{4} + \frac{\sqrt{97}}{4}$ and $x = -\frac{3}{4} + \frac{\sqrt{105}}{4}$

$\therefore \left\{x: x \in \mathbb{R}, -\frac{3}{4} + \frac{\sqrt{105}}{4} < x < -\frac{1}{4} + \frac{\sqrt{97}}{4}\right\}$

4) a)



4) b)

$x^3 + x^2 - 4x - 4 = 10x + 20 \Rightarrow x^3 + x^2 - 14x - 24 = 0 \Rightarrow (x-4)(x+2)(x+3) = 0$

$x = -3$ or $x = -2$ or $x = 4$

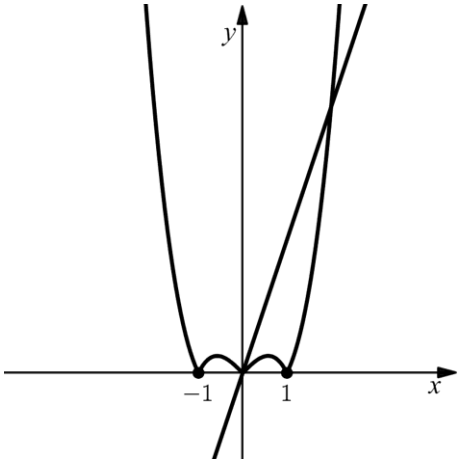
So $x = -2$ or $x = 4$

$\therefore x \in (-\infty, -2) \cup (4, \infty)$

5) a)

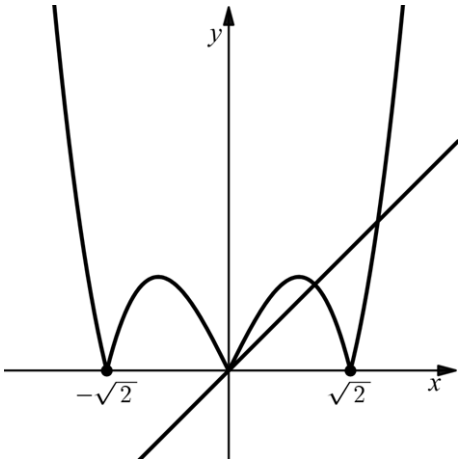
$x^3 - x = 3x \Rightarrow x^3 - 4x = 0 \Rightarrow x(x-2)(x+2) = 0$ so $x = -2$, $x = 0$ or $x = 2$

$x^3 - x = -3x \Rightarrow x^3 + 2x = 1 \Rightarrow x(x^2+2) = 0$ so $x = 0$



$\therefore \{x: x \in \mathbb{R}, x < 0\} \cup \{x: x \in \mathbb{R}, x > 2\}$

5) b)



$x^3 - 2x = x \Rightarrow x^3 - 3x = 0 \Rightarrow x(x^2-3) = 0$ so $x = 0$ or $x = \pm\sqrt{3}$

$x^3 - 2x = -x \Rightarrow x^3 - x = 0 \Rightarrow x(x^2-1) = 0$ so $x = 0$ or $x = \pm 1$

$\therefore \{x: x \in \mathbb{R}, x < 0\} \cup \{x: x \in \mathbb{R}, 0 < x < 1\} \cup \{x: x \in \mathbb{R}, x > \sqrt{3}\}$

5) c)

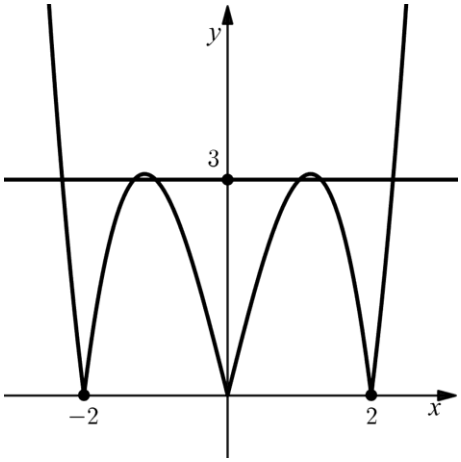
$x^3 - 4x = x(x^2 - 4) = x(x-2)(x+2)$

$x^3 - 4x = 3 \Rightarrow x^3 - 4x - 3 = 0 \Rightarrow (x+1)(x^2 - x - 3) = 0$

so $x = -1$ or $x = \frac{1}{2} \pm \frac{\sqrt{13}}{2}$

$x^3 - 4x = -3 \Rightarrow x^3 - 4x + 3 = 0 \Rightarrow (x-1)(x^2 + x - 3) = 0$

so $x = 1$ or $x = -\frac{1}{2} \pm \frac{\sqrt{13}}{2}$



$\therefore \left\{x: x \in \mathbb{R}, -\frac{1}{2} - \frac{\sqrt{13}}{2} \leq x \leq \frac{1}{2} - \frac{\sqrt{13}}{2}\right\} \cup \{x: x \in \mathbb{R}, -1 \leq x \leq 1\} \cup \left\{x: x \in \mathbb{R}, -\frac{1}{2} + \frac{\sqrt{13}}{2} \leq x \leq \frac{1}{2} + \frac{\sqrt{13}}{2}\right\}$

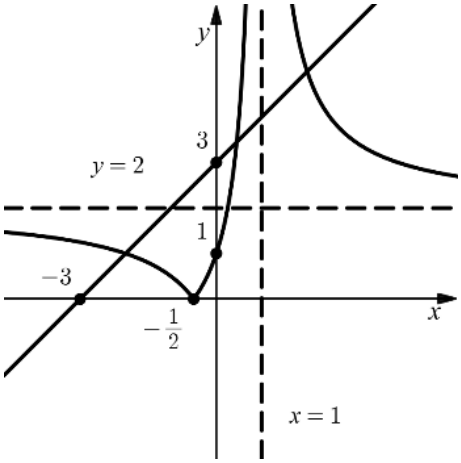
6) a)

Consider $y = \frac{2x+1}{x-1}$:

When $x = 0$, $y = -1$, when $y = 0$, $x = -\frac{1}{2}$

Asymptotes: $x = 1$, $y = 2$

Sketch $y = \left|\frac{2x+1}{x-1}\right|$ and $y = x + 3$



6) b)

$\frac{2x+1}{x-1} = x+3 \Rightarrow 2x+1 = (x+3)(x-1) \Rightarrow x^2-4=0 \Rightarrow x=-2$ or $x=2$

$\frac{2x+1}{x-1} = -(x+3) \Rightarrow 2x+1 = -(x+3)(x-1) \Rightarrow x^2+4x-2=0 \Rightarrow x=-2\pm\sqrt{6}$ so $x=-2+\sqrt{6}$

$\therefore \{x: x\in\mathbb{R}, x<-2\} \cup \{x: x\in\mathbb{R}, -2+\sqrt{6} < x < 1\} \cup \{x: x\in\mathbb{R}, 1 < x < 2\}$

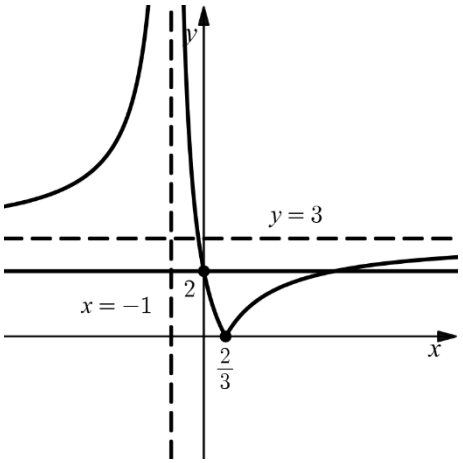
7) a)

Consider $y=\frac{3x-2}{x+1}$:

When $x=0$, $y=-2$, when $y=0$, $x=\frac{2}{3}$

Asymptotes: $x=-1$, $y=3$

Sketch $y=\left|\frac{3x-2}{x+1}\right|$ and $y=2$



$\frac{3x-2}{x+1} = 2 \Rightarrow 3x-2=2(x+1) \Rightarrow x=4$

$\therefore \{x: x\in\mathbb{R},\ x<-1\} \cup \{x: x\in\mathbb{R}, -1 < x < 0\} \cup \{x: x\in\mathbb{R}, x>4\}$

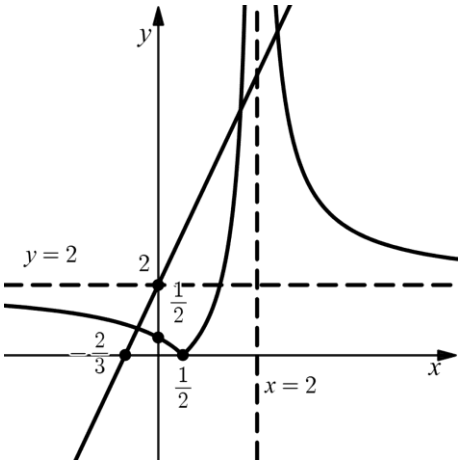
7) b)

Consider $y = \frac{2x-1}{x-2}$:

When $x = 0$, $y = \frac{1}{2}$, when $y = 0$, $x = \frac{1}{2}$

Asymptotes: $x = 2$, $y = 2$

Sketch $y = \left|\frac{2x-1}{x-2}\right|$ and $y = 3x + 2$



$\frac{2x-1}{x-2} = 3x + 2 \Rightarrow 2x - 1 = (3x + 2)(x - 2) \Rightarrow 3x^2 - 6x - 3 = 0$

$\Rightarrow x = 1 + \sqrt{2}$ or $x = 1 - \sqrt{2}$

$\frac{2x-1}{x-2} = -(3x + 2) \Rightarrow 2x - 1 = -(3x + 2)(x - 2) \Rightarrow 3x^2 - 2x - 5 = 0$

$\Rightarrow x = -1$ or $x = \frac{5}{3}$ so $x = \frac{5}{3}$

$\therefore \left\{x: x \in \mathbb{R}, 1 - \sqrt{2} < x < \frac{5}{3}\right\} \cup \left\{x: x \in \mathbb{R}, x > 1 + \sqrt{2}\right\}$

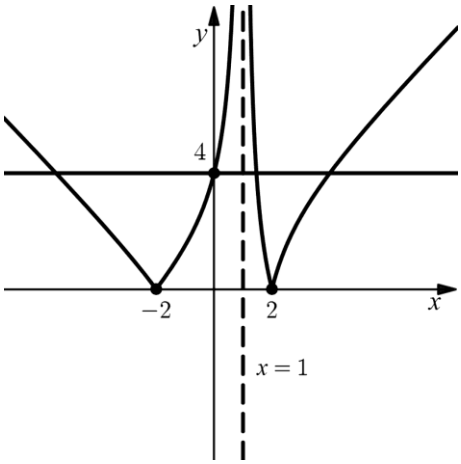
7) c)

Consider $y = \frac{x^2-4}{x-1}$:

When $x = 0$, $y = 4$, when $y = 0$, $x = \pm 2$

Asymptotes: $x = 1$

Sketch $y = \left|\frac{x^2-4}{x-1}\right|$ and $y = 4$

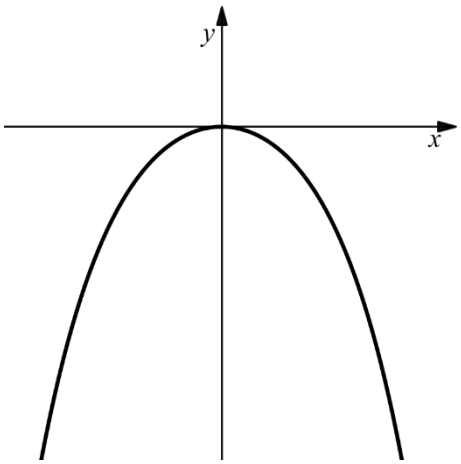


$\frac{x^2-4}{x-1} = 4 \Rightarrow x^2 - 4 = 4(x-1) \Rightarrow x^2 - 4x = 0 \Rightarrow x = 0$ or $x = 4$

$\frac{x^2-4}{x-1} = -4 \Rightarrow x^2 - 4 = -4(x-1) \Rightarrow x^2 + 4x - 8 = 0 \Rightarrow x = -2 \pm 2\sqrt{3}$

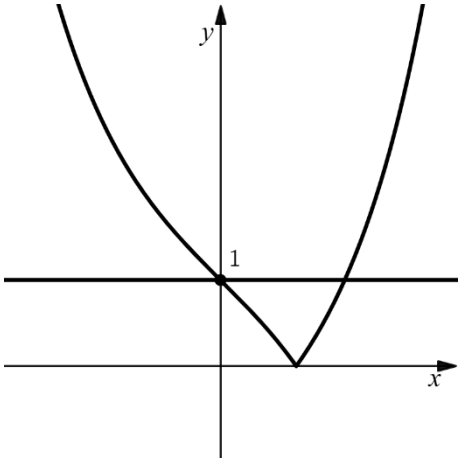
$\therefore \{x: x \in \mathbb{R}, -2 - 2\sqrt{3} \leq x \leq 0\} \cup \{x: x \in \mathbb{R}, -2 + 2\sqrt{3} \leq x \leq 4\}$

8) a)



$x \in (-\infty, 0) \cup (0, \infty)$

8) b)



$1 - \sinh x = 1 \Rightarrow \sinh x = 0 \Rightarrow x = \ln\left(0 + \sqrt{0^2 + 1}\right) = 0$

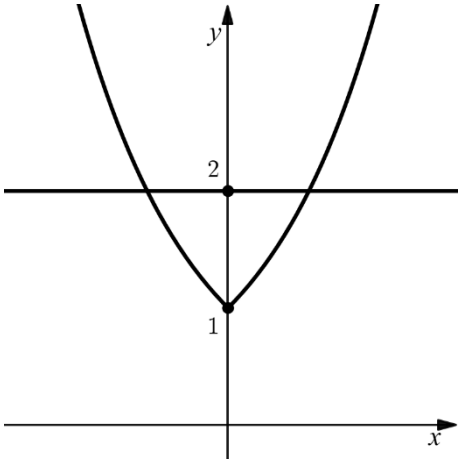
$1 - \sinh x = -1 \Rightarrow \sinh x = 2 \Rightarrow x = \ln\left(2 + \sqrt{2^2 + 1}\right) = \ln\left(2 + \sqrt{5}\right)$

$x \in \left[-1, \ln\left(2 + \sqrt{5}\right)\right]$

8) c)

$\sinh x + \cosh x = 2 \Rightarrow \frac{1}{2}(e^x - e^{-x}) + \frac{1}{2}(e^x + e^{-x}) = 2 \Rightarrow e^x = 2$ so $x = \ln 2$

$-\sinh x + \cosh x = 2 \Rightarrow -\frac{1}{2}(e^x - e^{-x}) + \frac{1}{2}(e^x + e^{-x}) = 2 \Rightarrow e^{-x} = 2$ so no solutions.



Then by symmetry, $x = -\ln 2$

$\therefore x \in (-\infty, -\ln 2] \cup [\ln 2, \infty)$

9)

$|-1 + \ln x| \leq 2$

$-1 + \ln x = 2 \Rightarrow \ln x = 3 \Rightarrow x = e^3$

$-1 + \ln x = -2 \Rightarrow \ln x = -1 \Rightarrow x = e^{-1}$

$\therefore \{x : x \in \mathbb{R}, e^{-1} \leq x \leq e^3\}$

10)

$\left|\frac{1}{x}\right| \geq 1$

$\frac{1}{x} = 1 \Rightarrow x = 1$

$\frac{1}{x} = -1 \Rightarrow x = -1$

$\therefore \{x: x \in \mathbb{R}, -1 \leq x < 0\} \cup \{x: x \in \mathbb{R}, 0 < x \leq 1\}$

11)

$\left|\cos x - \frac{1}{2}\right| \geq \frac{1}{2}$

$\cos x - \frac{1}{2} = \frac{1}{2} \Rightarrow \cos x = 1 \Rightarrow x = 0$

$\cos x - \frac{1}{2} = -\frac{1}{2} \Rightarrow \cos x = 0 \Rightarrow x = -\frac{\pi}{2}$ or $x = \frac{\pi}{2}$

$x \in \left(-\pi, -\frac{\pi}{2}\right] \cup [0] \cup \left[\frac{\pi}{2}, \pi\right]$

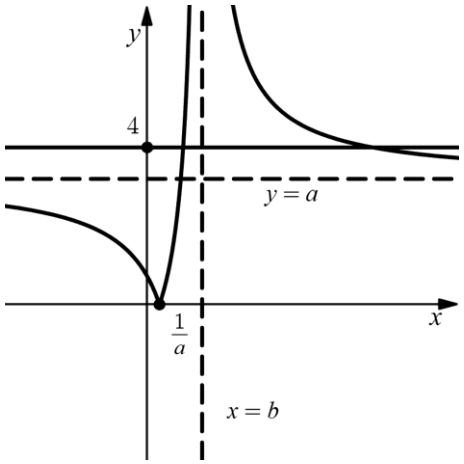
12)

Consider $y = \frac{ax-1}{x-b}$:

When $x = 0$, $y = \frac{1}{b}$, when $y = 0$, $x = \frac{1}{a}$

Asymptotes: $x = b, y = a$

Sketch $y = \left|\frac{ax-1}{x-b}\right|$ and $y = 4$



$\frac{ax-1}{x-b} = 4 \Rightarrow ax - 1 = 4(x - b) \Rightarrow ax - 1 = 4x - 4b \Rightarrow ax - 4x = 1 - 4b$

$\Rightarrow x = \frac{1-4b}{a-4}$

$\frac{ax-1}{x-b} = -4 \Rightarrow ax - 1 = -4(x - b) \Rightarrow ax - 1 = 4b - 4x \Rightarrow ax + 4x = 1 + 4b$

$\Rightarrow x = \frac{1+4b}{a+4}$

$\therefore \left\{x: x \in \mathbb{R}, \frac{1+4b}{a+4} \leq x < b\right\} \cup \left\{x: x \in \mathbb{R}, b < x \leq \frac{1-4b}{a-4}\right\}$